

CLOUD-BASED IMAGE PROCESSING SYSTEM

Image Upload & Automatic Thumbnail Generation

PHP

Amazon S3

AWS Lambda

Python

Pillow

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PROBLEM STATEMENT

The Problem

Uploading and processing images manually requires additional server-side work — and that gets harder to sustain as usage grows.

- 1 Increases server workload
- 2 Requires additional processing logic
- 3 Slows down the user-facing application
- 4 Becomes difficult to scale with many uploads

NEED

A system that can automatically generate thumbnails whenever an image is uploaded — without the main PHP application performing the image processing itself.

Upload once → processing happens automatically.

PROPOSED SOLUTION

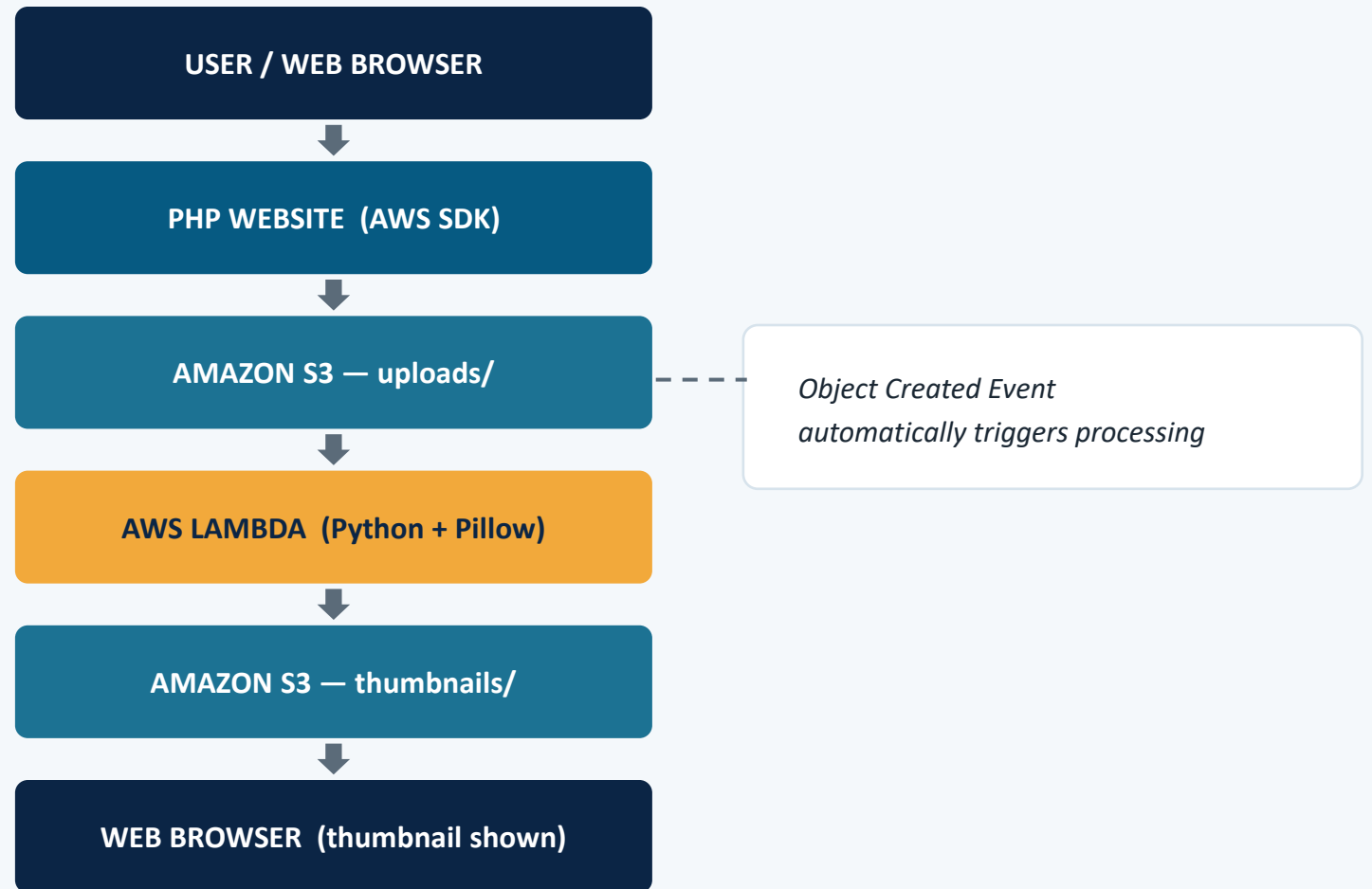
Our Idea

A cloud-based automatic thumbnail generation system — the user uploads an image through a PHP website, and everything else happens on its own.

- 1 Uploads the original image to Amazon S3
- 2 Detects the new upload
- 3 Triggers AWS Lambda
- 4 Processes the image using Python + Pillow
- 5 Generates a smaller thumbnail
- 6 Stores the thumbnail back in S3
- 7 Displays the generated thumbnail on the website

KEY IDEA: Upload once → Processing happens automatically

Proposed Architecture



How It Works

1

Select

User selects an image from the browser.

2

Validate & Upload

PHP validates the image and uploads it to S3.

3

Detect Event

S3 detects the new image via an Object Created Event.

4

Trigger Lambda

The event automatically triggers AWS Lambda.

5

Resize

Lambda downloads the image and uses Pillow to resize it.

6

Store Thumbnail

The thumbnail is uploaded to thumbnails/ in S3.

7

Display

The website checks for and displays the thumbnail.

Organized Image Storage

A single S3 bucket, image-thumbnail-bucket, contains two separate prefixes.

uploads/

- image1.jpg
- image2.png
- image3.webp

Lambda trigger is configured only for this prefix.



thumbnails/

- image1_thumb.jpg
- image2_thumb.jpg
- image3_thumb.jpg

Generated by Lambda — never re-triggers itself.

Separating prefixes prevents an infinite processing loop — writes to thumbnails/ never re-fire the uploads/ trigger.

Technologies Used

Technology	Purpose
HTML/CSS	User interface
JavaScript	Upload interaction and status checking
PHP	Backend and S3 communication
AWS SDK for PHP	Connects PHP with AWS
Amazon S3	Stores original images and thumbnails
AWS Lambda	Automatically processes uploaded images
Python	Lambda programming language
Pillow	Image resizing and thumbnail generation
AWS IAM	Access control and permissions
CloudWatch	Lambda monitoring and logs

KEY FEATURES

Main Features



Image upload through a web interface



Support for JPG, JPEG, PNG, GIF and WEBP



File type and size validation



Unique filenames for uploaded images



Automatic S3 storage



Event-driven Lambda execution



Automatic thumbnail generation



Aspect-ratio preservation



Separate original and thumbnail storage



Asynchronous thumbnail processing



Thumbnail status checking



Secure AWS credential handling



Private S3 storage with temporary access URLs

ADVANTAGES

Why This Approach?



Automatic

Thumbnail generation starts automatically after upload.



Cloud-Based

Images and processing use AWS cloud services.



Scalable

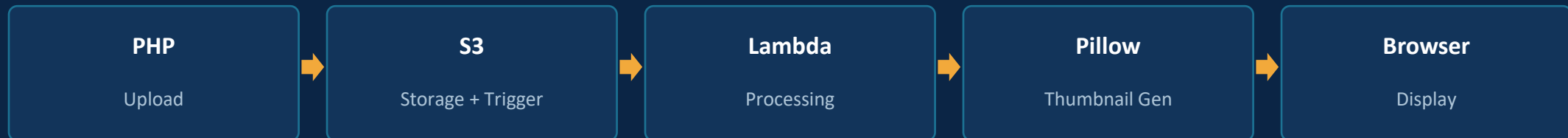
Lambda processes images without a continuously running server.



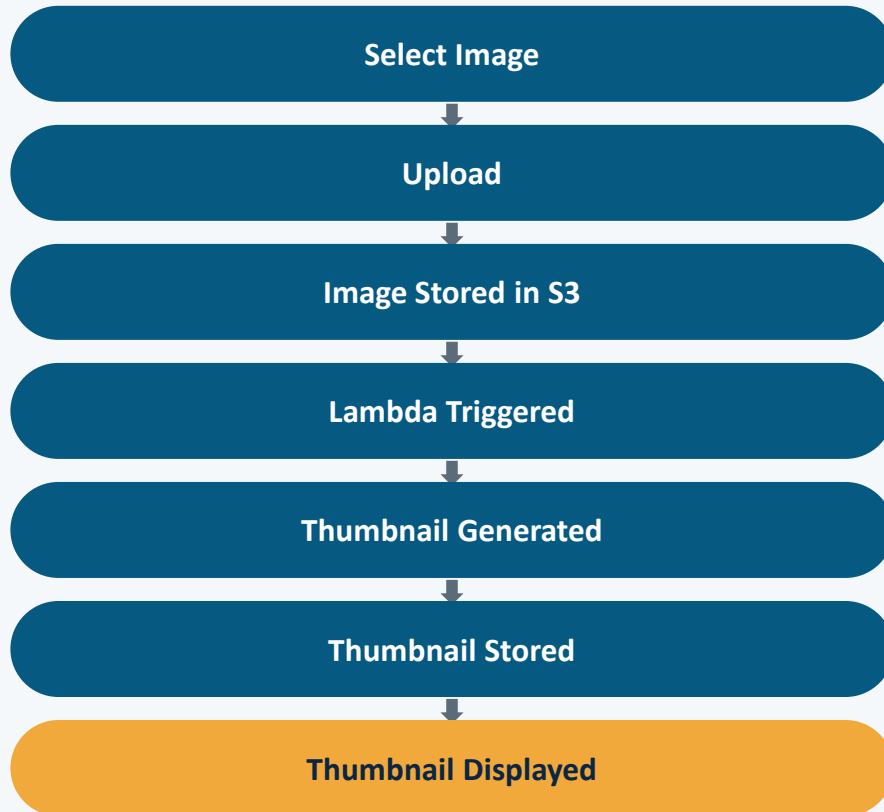
Secure

IAM permissions and private S3 storage control access.

MODULAR — Each component has a separate responsibility



From Upload to Thumbnail



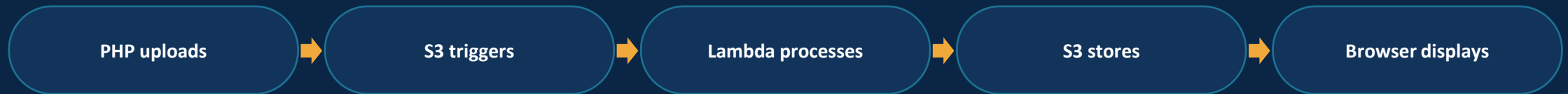
FUTURE SCOPE

Multiple image sizes	Batch image processing
Image compression	WebP/AVIF thumbnails
Metadata extraction	Image optimization for web
Upload history	User authentication
Progress tracking	Integration with larger systems

CONCLUSION

Automated, Cloud-Native Image Processing

The proposed system combines a traditional PHP web application with AWS cloud services to create an automated image-processing workflow, instead of processing images directly inside PHP.



This demonstrates cloud storage, serverless computing, event-driven architecture, and automated image processing in a single practical application.

Thank You

Questions?